Fiscal Year:	FY 2005	**Task Last Updated:**	FY 06/14/2006
PI Name:	Poirier, David Sc.D.		
Project Title:	Comparison of Structure and Segregation in Alloys Directionally Solidified in Terrestrial and Microgravity Environments		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Basics of solidification		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	85721-0012	**Congressional District:**	7
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	00-HEDS-02
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No. of Post Docs:	8	**No. of PhD Degrees:**	1
No. of PhD Candidates:	6	**No. of Master' Degrees:**	0
No. of Master's Candidates:	5	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	8	**Monitoring Center:**	NASA MSFC
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 6/01/2005 in June 2006 per PI (original end date was 6/01/2006)--jp 2/2009		
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):	Heinrich, J. C. (University of Arizona) Tewari, S. N. (Cleveland State University)		
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Task Description:	Terrestrial solidification experiments on/ Pb-Sb alloys have been done using different growth conditions to effect different dendritic microstructures. Numerical modeling to simulate the transport phenomena and solidification of the planned experiments is currently underway at both MSFC and the University of Arizona (UA). Terrestrial solidification experiments are underway at Cleveland State University (CSU). As ingot-samples are prepared at CSU, they will be characterized for dendrite morphology, macrosegregation, and microsegregation. The researchers at UA are also simulating dendritic growth in the presence of convection. This is a program to conduct experiments in a long duration microgravity environment, with the objective of studying dendritic microstructures and segregation in directionally solidified dendritic alloys. The research builds on our previous grant in which we generated an extensive data-base on directionally solidified Pb-Sn alloys and simulated solidification of the alloys using continuum theory of porous media. Based on our previous research, it is clear that thermosolutal convection masks the effects of diffusional transport during solidification and also leads to macro-segregates known as freckles. We are convinced, therefore, that with terrestrial experiments, diffusive and convective phenomena occur simultaneously, which complicate the study of dendritic growth. In microgravity, however, it would be possible to effect directional solidification with no thermosolutal convection. Our hypotheses are the convection and attendant macrosegregation will be eliminated and the dendritic microstructure will be greatly altered in microgravity. Models of dendritic solidification of alloys rely on the assumption of diffusional transports of both heat and solute. However, the thermosolutal convection of liquid in, from, or close to the solid plus liquid region of solidifying alloys masks the diffusional processes at the dendrite tips and is responsible for most macrosegregation in castings and ingots. Availability of microgravity provides an opportunity to obtain experimental data, where thermosolutal convection is negligible. With this grant, growth conditions for such microgravity experiments will be defined by combining mathematical modeling and simulations with the results of terrestrial experiments. Thermosolutal convection will be significantly reduced in microgravity, and thus we can compare segregation and microstructure in specimens terrestrially solidified and those solidified in microgravity. In terrestrial experiments, dendritic solidification is difficult to study because specimens are plagued with severe forms of macro segregation caused by thermosolutal convection, including: 1) continuous segregation from end-to-end; 2) occurrence of localized segregates referred to as freckles; and, 3) dendrite clustering. In the microgravity environment, the thermosolutal convection will be greatly diminished, and the convection will be confined mainly to the minor flows of interdendritic liquid required to satisfy solidification shrinkage and possibly due to "g-jitter." Morphological features, metrics, and segregation in samples directionally solidified in an earth-based furnace and in samples grown in microgravity are expected to be quite different. The technological significance of the research is that it will provide scientific underpinning to our understanding and control of how single-crystal castings solidify. Single-crystal castings are critical components in high-temperature gas-turbine engines that are used in high-speed aircraft.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	A major task of the NASA-sponsored research has been the development of a computer simulator for the dendritic solidification of binary alloys. The simulator is used to model the alloys that are used in the experimental part of the program. Under the auspices of other sponsors, we extended the simulator to predict a defect known as macrosegregation in commercial (multi-component) alloys. This was only possible because, under the auspices of NASA grants, we had developed the simulator for binary alloys. Freckle-defects can also form in ingots of high-value aerospace alloys made in special melting processes, such as vacuum arc-remelting (VAR) and electro-slag remelting (ESR). Under a FAA program, ingot producers and engine companies effect research and development in the Specialty Metal Processing Consortium (SMPC), centered at Sandia National Laboratories. The member companies are Allvac-an Allengheny Technology Company, Carpenter Technology Corporation, General Electric Aircraft Engines, Precision Rolled Products, Pratt-Whitney, Rolls-Royce Corp., RMI-Titanium, Schultz Steel, Special Metals Corp., and Trimet. The research team at The University of Arizona has been simulating ingot solidification with the goal of understanding and then avoiding processing conditions that lead to the freckle-defects. Recently, we enhanced the computer simulator by adding features to account for the thermodynamics and transport phenomena of gas-forming elements in solidifying alloys. We have been able to do this by parlaying our knowledge-base and computer program gained under NASA grants with a grant from the National Science Foundation and a contract with Sandia Laboratories. The main goal has been to enable us to use the solidification simulator to mitigate the formation of a casting defect known as microporosity in casting alloys. During this grant year, we worked with a leading investment casting company, Hitchiner Casting Co., to simulate the formation of microporosity in a steel alloy that is used as automotive rocker arms. The Hitchiner-collaboration is through our involvement with the Materials Corridor Initiative (MCI) that leverages the partial support of DOE with the support of industry to tackle problems and research that pertain to the economic sustainability of the USA-Mexico border region. We are also in the middle of a program sponsored by the Air Force Research Laboratory and Honeywell Engines, Systems & Services (with Howmet Casting) on simulating the formation of porosity in Ni-base superalloy castings. Improvements to a casting's microstructural homogenity will promote its lifetime, efficiency, and safety.
Task Progress:	The research was designed to conduct both ground-based experiments and experiments in a long duration microgravity environment with the objective of studying dendritic microstructures and segregation in directionally solidified alloys. The research team completed the Science Requirements and was authorized to proceed to RDR for eventual flight experiments. Along with several other research efforts in the Materials Science program, flight experiments were never realized because of NASA’s change in its major initiatives. Although flight experiments were never realized, the ground-based research was very fruitful. The major finding was thermosolutal convection leads to macro-segregates, known as “freckles.” Without requisite long-duration space experiments, computer simulations were done that showed segregation could be avoided in a microgravity environment. Experiments also led to the hypothesis that dendritic microstructures in directionally solidified (DS) alloys would be enhanced if grown in the absence of thermosolutal convection. A solidification model and computer program for simulating DS were developed at the University of Arizona (UA). Terrestrial DS experiments using Pb-Sn and then Pb-Sb alloys were done at Cleveland State University (CSU). The ingot-samples were characterized for dendrite morphology, macrosegregation and microsegregation. The UA-researchers ran simulations of the DS process of the experimental alloys in the presence of gravity, in microgravity environments and in microgravity with realistic “g-jitter.” The latter was done when the program was in flight definition status and preparing for flight experiments.

	Major Impacts of Research Although the program was aborted before flight experiments were carried out, the ground-based research led to several contributions to solidification processing of alloys. The research has provided scientific underpinning to understanding and controlling the DS-process and its sister process for producing dendritic monocrystals. Both DS castings and monocrystal castings are critical components in high-temperature gas-turbine engines that are used to power high-speed aircraft. The research led to development of computer software for simulating solidification and defect-formation in castings and ingots. In other sponsored research (non-NASA), the original code was being modified so that the formation of porosity in castings can be predicted and mitigated. Critical experiments and detailed analyses of the microstructures in the directionally solidified ingot-samples, which were made at Cleveland State University (CSU), verified that thermosolutal convection can cause macrosegregation and growth conditions leading to primary dendrite arm spacings and arrangements with stochastic characteristics. The research findings were disseminated through publication in peer-reviewed journals, presentations at major conferences and industrial laboratories, and articles in conference proceedings.
Bibliography Type:	Description: (Last Updated: 06/06/2025)
Articles in Peer-reviewed Journals	Zhao P, Heinrich JC, Poirier DR. "Dendritic solidification of binary alloys with free and forced convection." International Journal for Numerical Methods in Fluids. 2005 May 23;49(3):233-66. http://dx.doi.org/10.1002/fld.988 , May-2005
Articles in Peer-reviewed Journals	Zhao P, Heinrich JC, Poirier DR. "Numerical modeling of fluid–particle interactions." Computer Methods in Applied Mechanics and Engineering. 2006 Aug 15:195(41-43):5780-96. http://dx.doi.org/10.1016/j.cma.2005.09.023 , Aug-2006
Papers from Meeting Proceedings	Zhao P, Heinrich JC, Poirier DR. "Effect of variable density in dendritic solidification." 42nd AIAA Aerospace Sciences Meeting, Reno, Nevada, Jan. 5-8, 2004. 42nd AIAA Aerospace Sciences Meeting, AIAA paper 2004-628, January 2004. , Jan-2004